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CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	6.67×10^{-11} N m^{2} kg^{-2}
Acceleration due to gravity	g	=	9.8 m s^{-2}
1 Atmosphere	Atm	=	1.00×10^{5} N m^{-2}
Boltzmann's constant	k	=	1.38×10^{-23} J K^{-1}
Density of water		=	1.00×10^{3} kg m^{-3}
Specific heat capacity of water		=	4200 J kg^{-1} K^{-1}
Specific latent heat of fusion of ice		=	3.34×10^{5} J kg^{-1}
Specific latent heat of vaporization of water		=	2.26×10^{6} J kg^{-1}
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Stefan-Boltzmann's constant	s	=	5.67×10^{-8} W m^{-2} K^{-4}
Speed of light in free space	c	=	3.00×10^{8} m s^{-1}

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SECTION A

Answer ALL questions.

You MUST write your answers in the spaces provided in this answer booklet.

1. (a) (i) Write down an equation which shows the relation between the resultant force F on a body, the time t for which the force acts and the change in momentum, Δp, of the body.

[1 mark]

(ii) Define the term 'impulse'.

[1 mark]

(b) A model rocket of initial mass 1.8 kg is fired vertically into the air. Its mass decreases at a constant rate of 0.25 kg s^{-1} as the fuel burns. The final mass of the rocket is 0.40 kg. The rocket rises to a height such that, during the flight, the acceleration due to gravity may be considered to have the constant value of 9.8 m s^{-2}.

Calculate the

(i) **initial weight** of the rocket

[1 mark]

(ii) final weight of the rocket

[1 mark]

(iii) time taken for the fuel to be burned.

[2 marks]

(c) The data shown in Table 1 shows the variation with time, t, of the upward force on the rocket during the first 4 seconds after firing.

TABLE 1: UPWARD FORCE ON ROCKET VS TIME ELAPSED

Force (F) / N	0.0	11.0	17.0	18.5	18.0	16.5	15.0	10.0
Time (t) / s	0.0	0.5	1.0	1.5	2.0	2.5	3.0	4.0

(i) On the grid on the page opposite, plot a graph of the upward Force (F)/N versus Time (t) / s. Draw the BEST curve through the points. **[3 marks]**

(ii) On the SAME graph draw a line to represent the variation with time of the **total** weight of the rocket while the fuel is burning. **[2 marks]**

(iii) From the graph, read off the time delay between the firing of the rocket and lift off. Write this value in the space below.

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[1 mark]

(iv) On the graph, shade the area that represents the change in momentum of the rocket during the first 4 seconds after the rocket is fired. **[1 mark]**

(v) Estimate the value for the change in momentum of the rocket.

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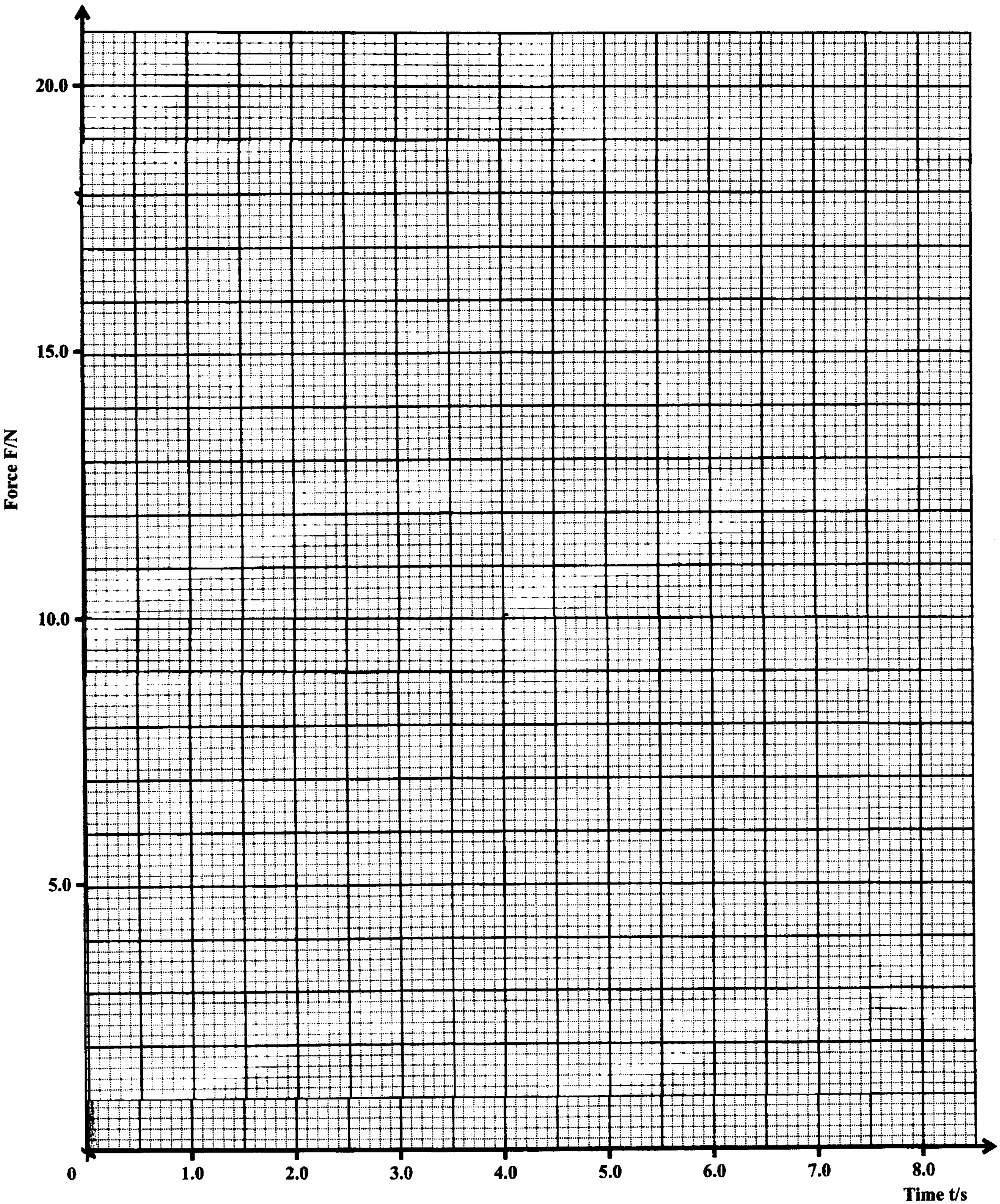
[2 marks]

Total 15 marks


Force F/N
20.0
15.0
10.0
5.0
0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
Time t/s

2. (a) Explain what is meant by

(i) 'periodic motion'

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[1 mark]

(ii) 'simple harmonic motion'.

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[2 marks]

(b) Figure 1 shows a graph of depth of water, h, against time, t, at a harbour.

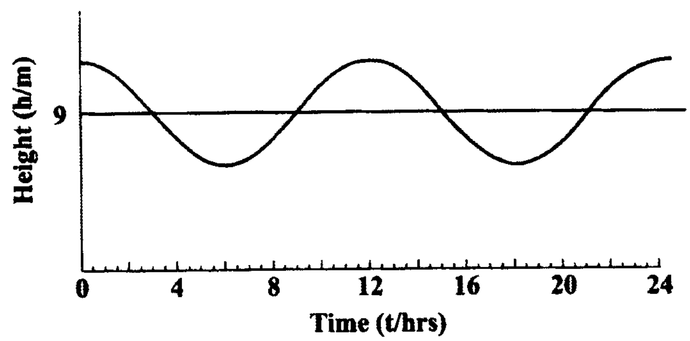

Figure 1

Describe the practical procedures that you would follow to obtain such a graph for this harbour.

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[5 marks]

(c) For a particular harbour, the variation of depth h of the water with time can be modelled by the equation

$$h = 9 + 5.0 \sin(\omega t)$$

where h is given in metres, t is the time in seconds and the constant ω is 1.45×10^{-4} rad s^{-1}.

For this harbour, calculate

(i) the MINIMUM depth of the water

[1 mark]

(ii) the two values of t, (t_1 and t_2), for which the water is 11.5 m deep. (Give the answer in hours)

Calculation of t_1	Calculation of t_2
$t_1 =$ hours	$t_2 =$ hours

[5 marks]

(iii) the length of time for EACH tide during which the depth of water is MORE than 11.5 metres.

[1 mark]

Total 15 marks

3. (a) A constant volume gas thermometer was calibrated at the ice point and at the steam point. The thermometer was then used to measure the temperature of a beaker of hot water. The readings for the height difference, h, between the mercury level in the closed limb and the open limb were –50.0 mm for ice point, +220 mm for the steam point and +105 mm for the temperature of the hot water.

Calculate the centigrade temperature of the hot water.

[3 marks]

(b) Temperature on the thermodynamic scale may be obtained using a constant volume gas thermometer. The thermodynamic temperature, T, is given by

$$\frac{T}{T_{tr}} = \frac{P_T}{P_{tr}},$$

extrapolated to zero pressure. Here, T_{tr} is the thermodynamic temperature of the triple point of water and P_T and P_{tr} are the pressures of the gas in the thermometer at temperature T and the triple point respectively.

(i) The value of T_{tr} is chosen to be 273.16 K. Why is this particular number used?

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[1 mark]

(ii) The bulb of a constant volume gas thermometer was maintained at the constant temperature T of a boiling liquid, and then at the triple point of water. Table 2 shows the readings taken for different masses of gas in the thermometer.

TABLE 2

P_T x 10^5 Pa	2.858	2.294	1.765	1.195	0.6598
P_{tr} x 10^5 Pa	4.337	3.480	2.677	1.812	1.000
P_T / P_{tr}					0.6598

a) Complete Table 2 by filling in the missing values of P_T/P_{tr}.

[1 mark]

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b) Give ONE reason why the values of P_T / P_{tr} are NOT constant.

[1 mark]

(c) On the grid provided on page 9, plot a **graph** of P_T / P_{tr} against P_{tr} starting the P_{tr} scale at 0. **[4 marks]**

(d) Read off the intercept on the $P_T \cdot P_{tr}$ **axis of your graph.**

[1 mark]

(e) Using your result in (d), deduce an accurate value of the thermodynamic temperature T.

[2 marks]

(f) Calculate the temperature found in Part (e) on the Celsius scale.

[1 mark]

Total 15 marks

SECTION B

Answer ALL questions.

You MUST write your answers in the separate answer booklet provided.

(a) (i) Define 'energy' and distinguish between 'kinetic energy' and 'gravitational potential energy'.

(ii) Show that the kinetic energy, E_K, of a body of mass, m, moving with speed, v, is given by the expression

$$E_K = \frac{1}{2} m v^2$$

[7 marks]

(b) Figure 2 shows a gymnast of mass 75.0 kg swinging from rest at a point P on a light rope to a point Q. The circular arc through his centre of mass has a radius of 10.0 m.

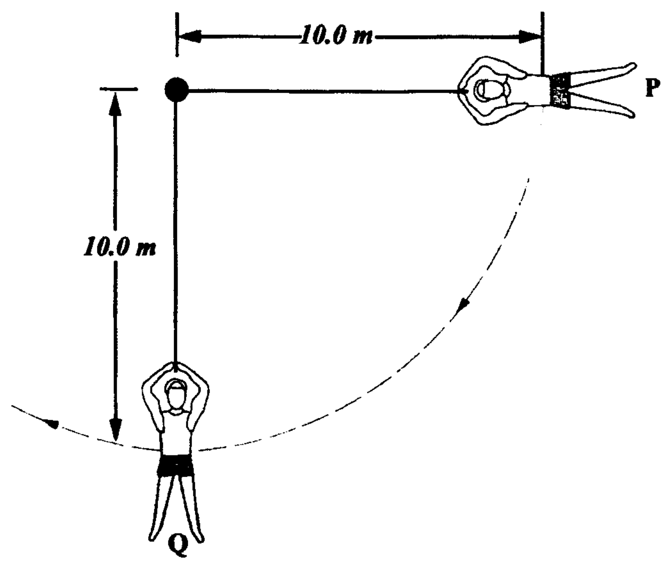

Figure 2

Calculate

(i) the gymnast's speed at Q

(ii) the angular velocity of the gymnast at Q. **[4 marks]**

(c) (i) Draw a free body diagram to show the forces acting on the gymnast at Q.

(ii) Calculate the tension in the rope as the string passes Q. **[4 marks]**

Total 15 marks

5. (a) Explain the following terms as they relate to the human eye:

(i) 'Accommodation'

(ii) 'Depth of focus' **[2 marks]**

(b) A student complains that she is not able to see clearly any object unless it is more than 80 cm from her eyes. The normal near point is taken as being 25 cm from the eye.

(i) Name the student's eye defect.

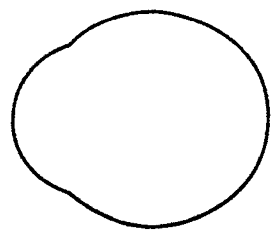

Figure 3

(ii) Copy Figure 3 onto your answer booklet. On the copied diagram, draw a ray diagram to illustrate the paths of two light rays from a point object at the normal near point, showing how they would reach the retina of the student's eye.

(iii) Draw a second ray diagram to show how a lens may be used to correct the defect for an object set at 25 cm from the eye. **[5 marks]**

(c) At age 45, a person is fitted for reading glasses of power 2.0 D in order to read at 25 cm. By the time she reaches 55, she discovers herself holding her newspaper at a distance of 40 cm in order to see it clearly with her glasses on. Calculate

(i) the position of her near point at age 45

(ii) the position of her near point at age 55

(iii) the power of the lens of the reading glasses now required so that she can again read at 25 cm. **[8 marks]**

Total 15 marks

6. (a) (i) Write down an equation for the First Law of Thermodynamics clearly explaining the meaning of the symbols used.

(ii) Explain why the heat capacity of a gas at constant pressure, C_p, is greater than the heat capacity of the gas at constant volume, C_v. Write down an expression that relates the two quantities with the Universal Gas Constant R.

[6 marks]

(b) The graphs in Figure 4 show data concerning the pressure, volume and temperature of a fixed mass of gas. The gas has a molar heat capacity of $\frac{3}{2}R$ at constant volume.

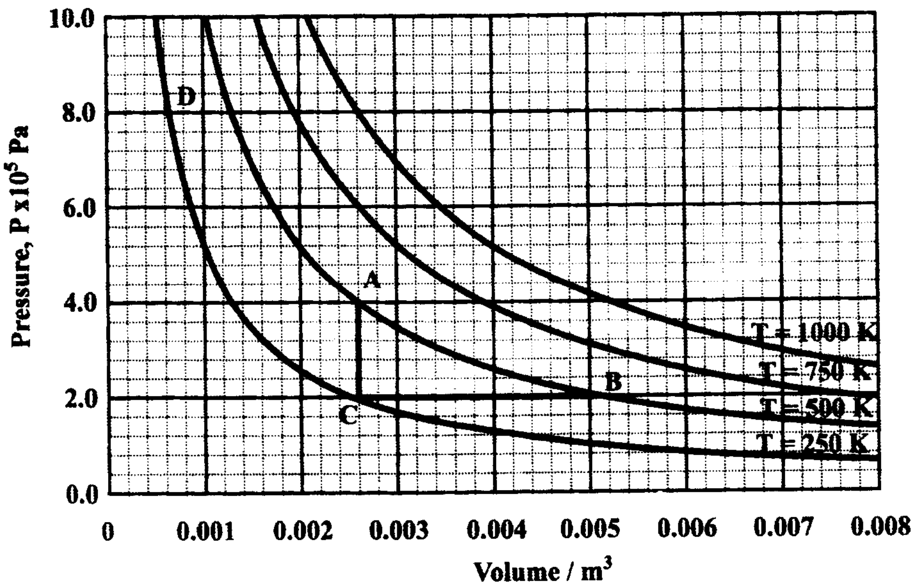

Figure 4

Using data from the **graph, calculate**

(i) the number of moles of gas

(ii) the quantity of heat required to take the gas

a) from C to A along CA

b) from C to B along CB

(iii) the work done by the gas along the path from C to B.

[9 marks]

Total 15 marks

END OF TEST